

# Casimir, CMB and $H_0$ — the shared $\xi$ scale

Step-by-step derivation of the connection formula  
with a consistency check of the SI conversion

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## What this is about

Three phenomena that at first have nothing to do with one another: the *Casimir effect* (vacuum force between plates,  $\sim \mu\text{m}$ ), the *cosmic microwave background* (CMB,  $T_{\text{CMB}} = 2.725\text{ K}$ ) and the *Hubble constant*  $H_0$  (cosmic expansion rate). In FFGFT all three hang on *one* vacuum length scale  $L_\xi \approx 100\ \mu\text{m}$  and on the same exponent  $41/4$  of the geometry parameter  $\xi = 4/30000$ . This note lays the connection open step by step (brought together in Doc. 166 from Doc. 091, 165, 025/061; status assessment per Doc. 190).

Throughout:  $\xi = \frac{4}{30000} = 1.3333 \times 10^{-4}$ .

## .1 Preliminary remark: why “Hubble constant” is misleading

Before we use  $H_0$ , a terminological clarification – because the usual designation *Hubble constant* is, from the FFGFT view, misleading on two counts.

**First: not a constant.** Already in standard cosmology  $H_0$  is not constant but the *present-day value* of the time-dependent Hubble parameter  $H(t) = \dot{a}/a$ . “Constant” there means only spatially homogeneous, not fixed in time. The name is thus already an established imprecision there.

**Second (and sharper): in FFGFT there is no expansion rate at all.** The static  $\xi$  universe has no scale factor  $a(t)$  and no time-varying  $H(t)$ ; the redshift is fractal path lengthening, not an expansion effect (Doc. 190, R14a/R15). There is therefore no “rate” whose present-day value one could take as  $H_0$ . The expansion reading is a  $\Lambda\text{CDM}$  pipeline interpretation (R18), not an FFGFT statement.

**What remains physically is a length.** The robust, model-neutral quantity is the *Hubble length*

$$R_H = \frac{c}{H_0}, \quad (1)$$

and  $H_0 = c/R_H$  is merely this length expressed, via the SI conversion factor  $c$ , as an inverse time.  $R_H$  is a *reference length* – the upper scale of the cosmological torus sector – entirely analogous to the Planck length  $\ell_P$  or to the system-specific bounds  $R = \hbar/mc$ . It is *not* the size of the universe and *not* the observable horizon (the latter  $\approx 3 R_H$  in  $\Lambda\text{CDM}$ ); only a lower bound is measurable (Doc. 190, R36).

### Terminology convention

In this note the cosmic scale is conceived primarily as a *reference length*  $R_H$ , not as a “constant”  $H_0$ . Where  $H_0$  appears it is to be read as  $H_0 = c/R_H$  – an SI restatement of the reference length, not a fundamental constant of nature. This is consistent with  $H_0$  as an emergent, decompactified quantity (R16) and with the status of  $R_H$  as a shared empirical reference point (R39). Precisely for this reason the following derivation works throughout with *lengths* ( $L_\xi$ ,  $R_H$ ) and carries  $H_0$  only as their conversion.

## .2 Starting point: three relations from three documents

### The three building blocks

1. **Casimir/vacuum (Doc. 091):** the CMB energy density as a T0 vacuum density,

$$\rho_{\text{CMB}} = \frac{\xi \hbar c}{L_\xi^4}. \quad (2)$$

2.  $H_0$  **ratio (Doc. 165):**

$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \xi^{10}. \quad (3)$$

3. **CMB temperature (Doc. 025/061):**  $T_{\text{CMB}} = 2.72548 \text{ K}$ , brought in via the Stefan-Boltzmann law (Step 1).

Relations (2) and (3) were developed independently. The question: are they two aspects of the same  $\xi$  structure?

## .3 Step 1: Casimir $\leftrightarrow$ CMB via the scale $L_\xi$

### 1a – Comparing the two energy densities

The standard Casimir energy density between two plates at separation  $d$  is

$$\rho_{\text{Casimir}}(d) = \frac{\pi^2 \hbar c}{720 d^4}. \quad (4)$$

The form  $\pi^2 \hbar c / (240 d^4)$  is the Casimir *pressure*, not the energy density; at  $d = L_\xi$  the energy density is  $4.29 \times 10^{-12} \text{ J/m}^3$ . At exactly the separation  $d = L_\xi$  it relates to the vacuum density (2) as

$$\frac{\rho_{\text{Casimir}}(L_\xi)}{\rho_{\text{CMB}}} = \frac{\pi^2 \hbar c / (720 L_\xi^4)}{\xi \hbar c / L_\xi^4} = \frac{\pi^2}{720 \xi} \approx 102.8 \quad (5)$$

Remarkable: inserting (2) into (5),  $\xi$  cancels and one recovers *exactly* the standard Casimir formula. The T0 notation is thus consistent with the textbook Casimir formula. The ratio (5) is therefore an identity, not a measurement:  $L_\xi$  is itself determined from the measured CMB energy density and  $\xi$  (Step 1b). At  $L_\xi$  the two densities do not coincide; equality would lie at a larger plate separation. The connection is a statement about scales, not evidence.

### 1b – $L_\xi$ from the CMB temperature

The CMB density is thermal (Stefan-Boltzmann):

$$\rho_{\text{CMB}} = \frac{\pi^2}{15} \frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3}. \quad (6)$$

Equating (2) and (6) and solving for  $L_\xi$ :

$$\frac{\xi \hbar c}{L_\xi^4} = \frac{\pi^2}{15} \frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3} \implies L_\xi = \left[ \frac{15 \xi}{\pi^2} \right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} \quad (7)$$

**Casimir exponent 1/4**

Numerically:  $L_\xi = 100.24 \mu\text{m}$  (0.11 % from the scale determined directly from  $\rho_{\text{CMB}}$ ).  
Decisive for what follows: from (7),  $\xi$  enters  $L_\xi$  with the power 1/4,

$$L_\xi \propto \xi^{1/4}.$$

**.4 Step 2:  $H_0$  as a  $\xi$  power**

Relation (3) solved for  $H_0/c$  (with the reduced Compton wavelength  $\bar{\lambda}_e = \hbar/m_e c$ ):

$$\frac{H_0}{c} = \frac{\pi}{2} \xi^{10} \frac{m_e c^2}{\hbar c} = \frac{\pi}{2} \frac{\xi^{10}}{\bar{\lambda}_e}. \quad (8)$$

 **$H_0$  exponent 10**

Inserted, this gives  $H_0 \approx 66.8 \text{ km/s/Mpc}$  – within 0.9 % of the Planck value 67.4 km/s/Mpc; the exponent 10 is set here, not derived forward (see the assessment below). Here  $\xi$  enters with the power 10:

$$H_0 \propto \xi^{10}.$$

**.5 Step 3: Multiplication – the bridge formula**

Now the product of the two scales  $L_\xi$  (7) and  $H_0/c$  (8). The powers of  $\xi$  add,  $\hbar c$  cancels:

$$\begin{aligned} L_\xi \cdot \frac{H_0}{c} &= \left[ \frac{15\xi}{\pi^2} \right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} \cdot \frac{\pi}{2} \frac{m_e c^2 \xi^{10}}{\hbar c} \\ &= \frac{\pi}{2} \left[ \frac{15}{\pi^2} \right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{1/4} \xi^{10} \\ &= \frac{\pi}{2} \left[ \frac{15}{\pi^2} \right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{41/4}. \end{aligned} \quad (9)$$

**Connection formula Casimir  $\leftrightarrow H_0$** 

$$L_\xi \cdot \frac{H_0}{c} = \underbrace{\frac{\pi}{2} \left( \frac{15}{\pi^2} \right)^{1/4}}_{K_{\text{geo}}=1.7441} \cdot \underbrace{\frac{m_e c^2}{k_B T_{\text{CMB}}}}_{=2.176 \times 10^9} \cdot \xi^{41/4}. \quad (10)$$

Numerical check:  $L_\xi \cdot H_0/c = 7.241 \times 10^{-31}$  – computed directly and via (10) algebraically identical (to  $10^{-14}$  %). The derivation is *exact* – no approximation, no free parameters *between* the building blocks.

## .6 The exponent $41/4$ and its decomposition

The exponent is the sum of the two individual exponents from Steps 1 and 2:

$$\frac{41}{4} = \underbrace{10}_{\substack{H_0 \text{ exponent} \\ (3)}} + \underbrace{\frac{1}{4}}_{\substack{\text{Casimir exponent} \\ (7)}} \quad (11)$$

The same exponent  $41/4$  appears independently in Doc. 026 as  $E_H = E_0 \xi^{41/4}$  (Hubble energy scale). The decomposition  $41/4 = 10 + 1/4$  is therefore not constructed by chance: the 10 carries the coupling electron mass  $\rightarrow$  Hubble scale, the  $1/4$  carries the Casimir scaling of the vacuum energy.

## .7 Where does the SI conversion factor sit?

A legitimate counter-question: we are operating here at the *decompactified* level of the formulation –  $H_0$  exists only after decompactification (Doc. 190, R16:  $H_0 = E_H/\hbar$  contains the  $\hbar$  that becomes real only then), and  $L_\xi$  is a *physical* length ( $\approx 100 \mu\text{m}$ ) built from  $\hbar$ ,  $c$ ,  $k_B$ . At this level an SI conversion factor *must* appear. It does – it merely sits hidden in two places, and in the exponent notation in a third.

### Place 1: the factor cancels (legitimate)

Both scales are built with the same  $\hbar c$ :

$$L_\xi \propto \frac{\hbar c}{k_B T_{\text{CMB}}}, \quad \frac{H_0}{c} = \frac{\pi}{2} \xi^{10} \frac{m_e c^2}{\hbar c}. \quad (12)$$

In the product  $\hbar c$  drops out – because  $L_\xi \cdot H_0/c$  is at its core a *length ratio* ( $L_\xi/\bar{\lambda}_e$  with the reduced Compton wavelength  $\bar{\lambda}_e = \hbar/m_e c$ ), and the ratio of two lengths is unit-free. This is no trick: the same SI conversion  $\hbar c$  builds *both* lengths, so it must cancel in the ratio.

### Place 2: the factor survives (the actual anchor)

What remains after the cancellation is *not* purely geometric:

$$\frac{m_e c^2}{k_B T_{\text{CMB}}} = 2.176 \times 10^9 \quad (13)$$

This contains  $c^2$  (mass  $\rightarrow$  energy) and  $k_B$  (Kelvin  $\rightarrow$  energy) – and it is a *ratio of two measured quantities*: electron rest energy to CMB temperature. *That* is the SI / empirical conversion factor. The bridge formula (10) is thereby geometric *form* ( $K_{\text{geo}} \xi^{41/4}$ ) **times** an SI anchor ( $m_e c^2/k_B T_{\text{CMB}}$ ) – *not* a pure geometric identity.

### Place 3: in the exponent – $41/4$ vs. $63/4$

The factor shows up most cleanly in the exponent itself. The  $41/4$  is referenced to the electron / CMB energy scale  $E_0$ , *not* to the Planck scale. Planck-referenced (Doc. 190, R35) the exponent is  $63/4$ , and

$$\frac{63}{4} = \frac{41}{4} + \frac{11}{2} \quad (14)$$

The  $11/2$  is exactly the SI scale conversion factor  $\ell_P \leftrightarrow E_0$  (Planck length vs. electron / CMB scale). Put differently: if one writes  $\xi^{41/4}$  and calls it “purely geometric”, one has tacitly absorbed  $\ell_P$  into the exponent – exactly the guardrail R35 (“the SI conversion factor  $\ell_P$  must not be absorbed into the exponent”). Which reference scale one makes explicit is a *choice*; the factor never disappears, it only changes location:

| Reference scale      | Exponent             | SI factor explicit as                |
|----------------------|----------------------|--------------------------------------|
| $E_0$ (electron/CMB) | $41/4$               | ratio $m_e c^2 / k_B T_{\text{CMB}}$ |
| $\ell_P$ (Planck)    | $63/4 = 41/4 + 11/2$ | extra exponent $11/2$                |

### Why we are necessarily decompactified here

This is not a matter of style:  $H_0$  *does not exist* in the compact  $T^4$  form – it contains the  $\hbar$  that becomes real only upon decompactification (R16) – and  $L_\xi$  is a concrete  $\mu\text{m}$  length that cannot be formulated without  $\hbar, c, k_B$ . The SI factors are therefore not optional but *constitutive*: they are what makes the quantities involved physical quantities in the first place. At the compact level there is neither  $H_0$  nor this bridge – the question of the SI factor does not even arise there.

### Verdict of the consistency check

The bridge formula is *consistent* – but as a *decompactified, SI-anchored* relation, not as a compactified  $T^4$  identity. The SI conversion factor appears exactly as expected: partly cancelled ( $\hbar c$ , legitimate as a length ratio), partly as the surviving empirical anchor ( $m_e c^2 / k_B T_{\text{CMB}}$ ), partly – in the exponent notation – as the  $11/2$  between the  $E_0$  and Planck references (14). This is fully in line with Doc. 190: R34 (SI conversions indispensable for the  $c$ -/time-power,  $c = \hbar = 1$  destroys the structure), R16 ( $H_0$  emergent/decompactified) and R35 (no absorption of  $\ell_P$  into the exponent).

## .8 Inversion: $H_0$ directly from the Casimir scale

Since  $L_\xi$  follows via (7) from  $\rho_{\text{CMB}}$  (i.e. from  $T_{\text{CMB}}$ ), (10) can be solved for  $H_0$ :

$$H_0 = \frac{\pi}{2} \left[ \frac{15}{\pi^2} \right]^{1/4} \frac{c}{L_\xi} \xi^{41/4} \quad (15)$$

This is the experimental lever (Doc. 166, §5): a *Casimir precision measurement* of the scale  $L_\xi$  fixes, via (15), a value for  $H_0$  – an  $H_0$  determination in the laboratory rather than in the sky. This presupposes that  $L_\xi$  has a physical Casimir signature of its own; that is open, since here  $L_\xi$  is determined from  $T_{\text{CMB}}$ .

## .9 Complete number chain

| Quantity                                    | Expression                                                 | Value                                 |
|---------------------------------------------|------------------------------------------------------------|---------------------------------------|
| $\xi$                                       | $4/30000$                                                  | $1.3333 \times 10^{-4}$               |
| $T_{\text{CMB}}$                            | Planck 2018                                                | 2.72548 K                             |
| $L_\xi$                                     | $[15\xi/\pi^2]^{1/4} \hbar c / (k_B T_{\text{CMB}})$       | 100.24 $\mu\text{m}$                  |
| $\rho_{\text{CMB}}$                         | $\xi \hbar c / L_\xi^4$                                    | $4.170 \times 10^{-14} \text{ J/m}^3$ |
| $\rho_{\text{Casimir}} / \rho_{\text{CMB}}$ | $\pi^2 / (720 \xi)$                                        | $\approx 102.8$                       |
| $K_{\text{geo}}$                            | $(\pi/2)(15/\pi^2)^{1/4}$                                  | 1.7441                                |
| $m_e c^2 / (k_B T_{\text{CMB}})$            | –                                                          | $2.176 \times 10^9$                   |
| $\xi^{41/4}$                                | $41/4 = 10.25$                                             | $1.91 \times 10^{-40}$                |
| $\hbar H_0 / (m_e c^2)$                     | $(\pi/2) \xi^{10}$                                         | $2.789 \times 10^{-39}$               |
| $L_\xi \cdot H_0 / c$                       | $K_{\text{geo}} (m_e c^2 / k_B T_{\text{CMB}}) \xi^{41/4}$ | $7.241 \times 10^{-31}$               |
| $H_0$                                       | (15)                                                       | 66.82 km/s/Mpc                        |

Cross-check:  $H_0 = 66.82 \text{ km/s/Mpc} = 2.166 \times 10^{-18} \text{ s}^{-1}$  gives  $\hbar H_0 / (m_e c^2) = 2.79 \times 10^{-39}$ .

## .10 Honest assessment of the status (Doc. 190)

The chain above is algebraically clean, but its *epistemic* status must remain clear (corrections register Doc. 190, entries R20, R32, R33, R35, R36, R39):

### What is derived – and what is not

- **Form forward:** that the connection is a *dimensionless  $\xi$  ratio* (Casimir scale  $\times$  Hubble scale =  $\xi$  power) is structurally forward. The *chaining* of the building blocks (9) is exact.
- **Exponent not forward (R20):** the value  $41/4$  – already the partial exponent 10 from (3) – is *not* derived forward from the  $T^4$  geometry. The neat decomposition  $41/4 = 10 + 1/4$  (11) is *structurally suggestive*, not a proof.
- **$\xi^N$  guardrail (R35):** “ $X = \xi^N$ ” counts as content only if  $N$  is derived forward; otherwise any magnitude can be written as a  $\xi$  power. The SI factor  $\ell_p$  must not be tacitly absorbed into the exponent (cf. Place 3 above).
- **$H_0$  is emergent, not fundamental (R16/R32):**  $H_0$  contains the  $\hbar$  that becomes real only upon decompactification and does not exist in the compact  $T^4$  form.
- **Shared reference point (R39):** the Planck length  $\leftrightarrow R_H$  relation follows from *no* framework out of first principles;  $\Lambda\text{CDM}$  fixes it via the measured  $H_0$ . FFGFT adopts the established value for comparability – this is *not* an asymmetric disadvantage but a shared empirical reference point of both models.

### Verdict of the assessment

The *chaining* is exact – a consistency relation among SI-anchored quantities ( $L_\xi$ ,  $R_H$ ,  $m_e$ ,  $T_{\text{CMB}}$ ), together with the possible experimental lever Casimir  $\rightarrow H_0$  (15), which presupposes a Casimir signature of  $L_\xi$  of its own (open).  $\ell_p$  does *not* appear in the formula; via the  $E_0$  reference it is *extrapolable* – as a consistency check, not a derivation: the leg  $E_0 \leftrightarrow \ell_p$  (11/2) coincides with the exponent of  $\alpha = \xi E_0^2$  (see



Outlook), whose agreement with the measured value is a property of the anchor  $E_0$ , not an independent prediction; the cosmic leg (41/4) remains open (R20). It is therefore *not* a first-principles prediction of  $H_0$ .

## .11 Outlook: the 11/2, the $\Lambda$ fine-tuning problem and the $H_0$ tension

**The 11/2 is a starting point, not a proof.** The exponent 11/2 in the split  $63/4 = 41/4 + 11/2$  is not arbitrary: it coincides with the fine-structure exponent,

$$\alpha = \xi \cdot E_0^2 = K \cdot \xi^{11/2}, \quad E_0 = \sqrt{m_e m_\mu} \propto \xi^{9/4} \quad (\text{Doc. 011, 070, 202, 210}), \quad (16)$$

and it is the *same*  $E_0$  to which the cosmic exponent is referenced ( $E_H = E_0 \xi^{41/4}$ , Doc. 026). The agreement of  $\alpha = \xi E_0^2$  with the measured value is a property of the anchor  $E_0$ , which provides the connection to the SI, not an independent prediction. Numerically  $E_0$  sits exactly where the split requires:  $E_0/E_{\text{Planck}} = \xi^{5.48}$  versus  $11/2 = 5.5$  – a match to  $\sim 0.5\%$  (precisely the rounding 15.72 vs. 15.75 from R35).

**However:** the  $\alpha$  exponent is referenced to the MeV / lepton base (the 9/4 scaling), the split 11/2 to the Planck scale (the offset) – different reference bases, the same number. Whether this is *structurally identical* or a numerical hit in the dense  $\xi^N$  field is open; the guardrail (R35, Doc. 271) requires for “content” an isolated, predicted, precise value. The identity would be proven only by  $E_0 = E_{\text{Planck}} \cdot \xi^{11/2}$  forward. *Recorded as a conjecture, not as evidence.*

**The reverse question: why does  $\Lambda$ CDM’s  $H_0$  sit there?** Were there a genuine, fixed connection  $\ell_P \leftrightarrow R_H$ , that would be a sharp statement about  $H_0$ . But  $\Lambda$ CDM does *not* choose  $H_0$  from theory – it *measures* it (and contested: the  $H_0$  tension  $\sim 67$  vs.  $\sim 73$ );  $\ell_P$  comes from the measured  $G, \hbar, c$ . The ratio is not harmless:

$$\frac{\ell_P}{R_H} \approx 1.2 \times 10^{-61}, \quad \left( \frac{\ell_P}{R_H} \right)^2 \approx 10^{-122} \sim \frac{\rho_\Lambda}{\rho_{\text{Planck}}}. \quad (17)$$

The  $(\ell_P/R_H)^2 \sim 10^{-122}$  is the cosmological constant problem – the largest unexplained fine-tuning in physics.  $\Lambda$ CDM “chooses” the value by fitting  $\Lambda$  to the observed acceleration, without a theoretical reason. FFGFT eliminates  $\Lambda$  (R17/R39) and does *not* inherit the fine-tuning – but relocates the same “why this scale” into the exponent 41/4.

**The test.** If the 41/4 became forward-derivable (together with the 11/2 leg), FFGFT would *predict*  $\ell_P/R_H$  – and hence  $H_0$  – instead of importing it; it could even decide the  $H_0$  tension (which side, 67 or 73). Precisely that would turn the conjecture into evidence (isolation/prediction/precision, Doc. 271). Until then it remains a suggestive restatement of a measured ratio, open.